Data Wrangling (INFOMDW)

Assignment 2

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Summary and experiment reproduction of the paper:
“MissForest - nonparametric missing value imputation for mixed-type data”

Summary of the paper

Problem and proposed method

The paper starts with introducing the main problem associated with data missingness, particularly in fields such as medicine and biology. In these research fields, data is complete and often contains missing values. With advent of new measurements techniques in these fields arises the problem of high dimensionality and mixed typed of data, which include both continuous and categorical variables. In addition, this kind of data holds complex and nonlinear relationships, which are hard to deal with using parametric imputation methods.
The paper mentions parametric imputation methods, such as KNNimpute, saturated multinomial model and MICE that require tuning of parameters, assumptions about the data distribution, and many of them only cover one type of variables. All these factors lead to the degradation in imputation performance and the method introduced in this paper tackles with all these downsides.

The paper "MissForest – nonparametric missing value imputation for mixed typed data introduces the novel method of imputing missing values in the dataset using a random Forest. This method works well on mixed and high-dimensional type of data and able to capture well non-linear relationships between variables. It is an iterative process, where a random forest is trained on the observed data and imputes missing values with its predictions. Imputation performance is evaluated using out of bag dataset, that didn't participate in the training, and therefore validation set is not required for this method. MissForest imputation does not require tuning parameters and does not make assumptions about data, which makes utilizing it simpler compared to similar methods, such as KNNimpute or MICE.

MissForest Iterative algorithm

The following notations were used in the paper:

Yobs/Ymis - the observed and missing values of variable X,

Xobs/Xmis - the other variables in the rows where X is observed/missing

The MissForest iterative imputation is performed according to the following steps:

- The missing values of variable X are guesses using a mean imputation. Then all the variables Xs, where s=1…p (p is number of features) are sorted according to the quantity of missing values in an ascending order. For each variable Xs, imputation is performed in two steps:
- Random forest is fit with values of Yobs and predictors Xobs.
- Random forest is applied on Xmis to predict Ymis values. The imputation continues until the stopping criterion is reached.

Stopping criterion for N continuous variables types is given by formula:

$$\Delta_N = \frac{\sum_{j \in \mathbf{N}} (\mathbf{X}_{new}^{imp} - \mathbf{X}_{old}^{imp})^2}{\sum_{j \in \mathbf{N}} (\mathbf{X}_{new}^{imp})^2},$$

, which compares imputed values between two consecutive rounds.

For categorical variables, stopping criterion is given by the formula:

$$\Delta_F = \frac{\sum_{j \in \mathbf{F}} \sum_{i=1}^{n} \mathbf{I}_{\mathbf{X}_{new}^{imp} \neq \mathbf{X}_{old}^{imp}}}{\#\mathrm{NA}},$$

, which compares categorical entries between rounds

At each iteration, MissForest refines imputation values, until the change between iterations stops decreasing. The stopping conditions are the criterions given above.

Imputation performance for continuous variables is measured with normalized root mean square error (NRMSE):

$$\text{NRMSE} = \sqrt{\frac{\text{mean}\left((\mathbf{X}^{true} - \mathbf{X}^{imp})^2\right)}{\text{var}\left(\mathbf{X}^{true}\right)}},$$

It is computed over the entries that were artificially removed during imputation. The normalization makes errors comparable across variables and datasets with different scales.

For categorical variables, imputation performance is calculated with falsely classified entries metric (PFC):

$$\text{PFC} = \frac{\text{number of wrongly imputed categorical entries}}{\text{total number of imputed categorical entries}}$$

It is the share of imputed categorical variables that do not match the true category. It is computed only over the entries that were removed.

Performance values for both metrics vary from 0 for good imputation to 1 for a poor one. An NRMSE of 1 means that imputation error is about as large as the natural spread of the variable. Lower values mean imputed values are closer to the true ones.

To predict the true imputation error, Out-of-bag (OOB) error estimates are used. Out of bag errors for each variable is generated during random forest fit on observed values. The performance of this estimation is measured by difference between true imputation error and OOB error estimate.

Strengths of MissForest:

- MissForest can perform imputation on any kind of data, it does not need tuning parameters or assumptions about data distribution.
- It outperforms established methods such KNN imputation and MICE and can assess the quality of imputation without needing test set.
- MissForest provides great imputations results for high dimension data with complex interactions and nonlinear relations.

Limitations of MissForest:

- Execution speed: it is much slower compared to KNN imputation.
- OOB estimates are not perfect, they underestimate the error for categorical data.
- Imputation accuracy is lower compared to other methods in some datasets.

The Evaluation methodology:

In the paper, MissForest imputation error rates are compared with other known imputation methods, such as KNNimpute and MICE on datasets with 10,20,30% data removed at random. NRMSE and PFC metrics are calculated depending on the variable type and Wilcoxon test is performed to compare error rates between MissForest and other methods. The datasets used in the comparison come from various biological and clinical fields, and were obtained from UCI machine learning repository and provided by original researchers.

For continuous data, MissForest reduced the NRMSE by up to 25% compared to KNNimpute, and by more than 50% on the Musk dataset, although MissPALasso performed slightly better on the Isoprenoid data. For categorical data, MissForest outperformed all other methods, reducing the PFC by up to 60% compared to MICE on the SPECT data and by 10–20% on the other datasets. For mixed-type data, it often reduced the imputation error by more than 50% compared to KNNImpute and MICE.

OOB error estimates were compared to true imputation errors for continuous and categorical datasets to evaluate how accurate were OOB error estimates. For continuous datasets, OOB estimates closely match true imputation errors, while for categorical datasets, OOB estimates underestimated imputation error. Overall, OOB estimates served as a reasonable approximation of true error estimates, and can be used to evaluate imputation quality.

Implementation of the MissForest imputation method

Discussion of the results

The MissForest imputation experiment on categorical data was reproduced in R and is available in a Github repository https://github.com/zangphan96/DW-assignment-2. This experiment was chosen because all of its datasets are publicly available and contain a manageable number of features compared to continuous and mixed-type datasets. As in the original study, three categorical datasets were used, all obtained from the UCI Machine Learning Repository:

- SPECT: cardiac single photon emission computed tomography data, with 22 binary features summarizing over 3,000 2D SPECT images from 267 patients
- Promoter: E. coli promoter gene sequences, with 106 sequences of 57 base pairs, each position taking one of four nucleotides, plus a variable distinguishing promoter from non-promoter instances
- Lymphography: 148 lymphoma patients described by 19 mostly nominal properties, nine of them binary

The imputation results obtained with the script missforest_categorical.R, averaged over 50 simulations for each dataset and missingness rate (10%, 20% and 30%), are given in Table 1 and Figure 1.

Dataset	Missing	Runs	PFC (true)	SE	PFC (OOB)	True - OOB	Time [s]	Paper time [s]
SPECT	10%	50	0.178	0.0024	0.161	0.016	5.6	5.5
SPECT	20%	50	0.187	0.0021	0.154	0.034	5.6	5.5
SPECT	30%	50	0.201	0.0013	0.142	0.058	5.0	5.5
Promoter	10%	50	0.669	0.0025	0.656	0.013	17.5	38.0
Promoter	20%	50	0.678	0.0015	0.636	0.041	20.7	38.0
Promoter	30%	50	0.681	0.0014	0.615	0.067	22.5	38.0
Lymphography	10%	50	0.318	0.0038	0.296	0.022	6.0	7.0
Lymphography	20%	50	0.339	0.0025	0.275	0.063	7.5	7.0
Lymphography	30%	50	0.347	0.0022	0.253	0.095	6.7	7.0

Table 1. Imputation results for datasets with different missingness rates

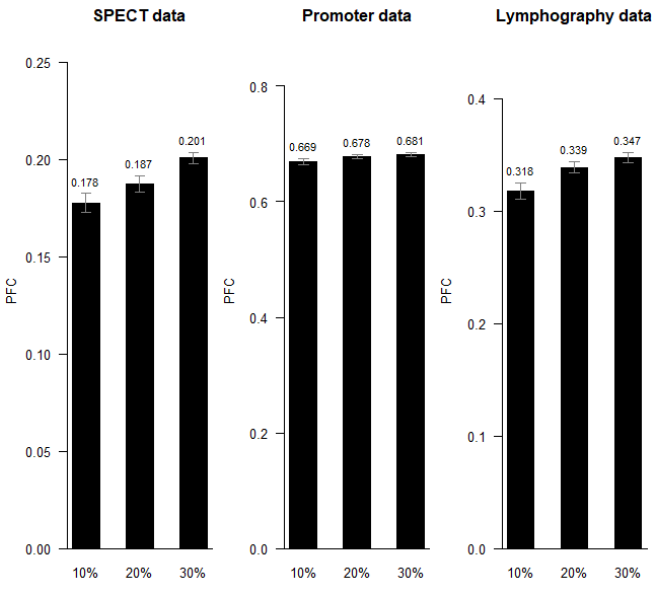


Figure 1. PFC values over datasets with missingness rates of 10 %, 20 %, 30 %

It can be seen from Figure 1 that PFC values are increasing with the missingness rate for all datasets, reflecting that imputed values predictions are less accurate with reduced information. The PFC values for all datasets closely resemble the ones shown in the paper's histogram and therefore the replication was successful. The standard errors received were in the order of 10^-3, slightly larger than 10^-4 reported in the paper. This is most likely attributed to different number of simulations runs, however both numbers are small and show that

PFC values were accurate across different simulations. The execution times for simulations in table closely resemble those shown in the paper. The execution times for promoter datasets are even halved compared to the paper's, which can be explained by a newer hardware used in the simulations.

Difference between true imputation errors and OOB estimates are shown in the Figure 2 and also written in the Table 1 under the column "True-OOB".

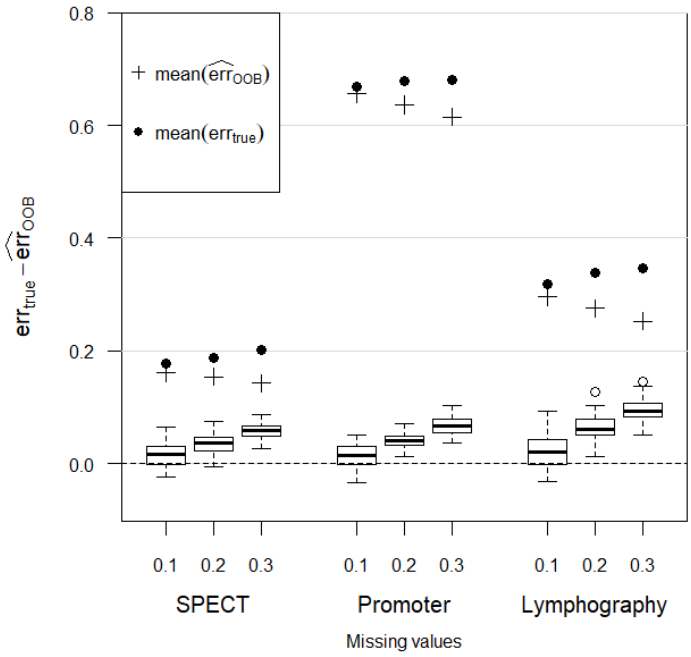


Figure 2. Differences between true imputation values and OOB estimates over datasets with missingness rates of 10 %, 20 %, 30 %

The difference values across all datasets and for different missingness values closely resemble those shown in the paper's box plots and therefore the reproduction was successful. The OOB estimates underestimate true imputation errors as shown by the mean markers and this underestimation grows with missingness rate. With higher missingness rate, there are more imputed values- these imputed values are later used in the imputation predictions and create a data leakage. Therefore, the OOB estimates become more positive with higher missingness rate.

Parameters used in the code

The script missforest_categorical.R was run in R Studio with the following MissForest parameters:

- N_SIM <- 50 - number of simulations
- MISS_RATES <- c(0.1,0.2,0.3) – data missingness rate configurations of 10%, 20% and 30%
- NTREE <-100 - number of decision trees in a random forest
- mtry <- sqrt(p), where p is the total number of features - the number of features randomly selected as candidates at each split
- MAXITER <-10 - maximum number of rounds allowed for imputations
- BACKEND <- "randomForest"- package used internally to build the tress
- N_CORES <-5 - number of processor cores for parallel simulations
- SEED <-2012 - fixed seed for reproducibility